

Advanced Modules for School Management System (Post-Completion)

1. Academic Planning & Timetable Management Module

Core real-world school requirement

Features

- Class-wise & teacher-wise timetable creation
- Conflict detection (teacher / classroom clashes)
- Subject period allocation
- Auto-generated weekly timetable
- Student & teacher timetable views

Advanced Concepts

- Scheduling algorithms
 - Constraint validation
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2. Online Examination & Assessment Module

Converts system into digital learning platform

Features

- Online exams (MCQ / subjective)
- Time-based exams
- Auto-evaluation for MCQs
- Manual evaluation for subjective answers
- Exam result & ranking system

Advanced Concepts

- Exam timers
- Secure submission
- Auto grading logic

3. Learning Management System (LMS) Module

Adds EdTech value

Features

- Upload video lectures & notes
- Chapter-wise content
- Assignment + learning material linking
- Student progress tracking
- Completion percentage

Advanced Concepts

- Media handling
- Progress analytics

4. Parent Portal Module

Most schools require this

Features

- Parent login (linked to student)
- View attendance, performance, fees
- Teacher remarks & feedback
- Download reports & receipts
- Notifications from school

Advanced Concepts

- Parent-student mapping
- Controlled data visibility

5. Smart Attendance System (Biometric / QR)

Automation & IoT-ready

Features

- QR-based student attendance
- Location & time validation
- Teacher attendance via QR / mobile

- Attendance fraud prevention

Advanced Concepts

- Token-based QR validation
- Time-window logic

6. Fee Automation & Accounting Module

Financial transparency

Features

- Online fee payment (Razorpay)
- Partial payments
- Late fee calculation
- Auto reminders
- Class-wise & month-wise fee reports

Advanced Concepts

- Transaction reconciliation
- Payment webhooks

7. Notification & Communication Module

Real-time engagement

Features

- Email / SMS / WhatsApp notifications
- Exam alerts
- Fee reminders
- Attendance alerts to parents
- Announcement broadcasting

Advanced Concepts

- Event-driven notifications
- Queue-based messaging

8. Analytics, AI & Performance Insights Module

Decision-making intelligence

Features

- Student performance prediction
- Attendance vs performance correlation
- Teacher performance analytics
- Drop-risk identification
- Downloadable analytics reports

Advanced Concepts

- Data analytics
- Basic ML logic